

the replication principle

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outline

the idea

get code from others!

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bad excel

- ◇ never trust numbers that come from excel
 - no way to find out what happened, there's no code!
[*]<http://andrewgelman.com/2013/04/17/excel-bashing/>
- ◇ I learned it hard way:
 - my first paper for ecological economics, done in excel
 - reviewers got back after 6mo, i had dozens of excel files
 - couldn't replicate my own results!
- ◇ “Talk is cheap. Show me the code”—Linus

replication, replication

- ◇ replication=write computer code that will do
everything
 - from raw data (eg FED, IMF) to results (eg regression)
- ◇ necessary for science
- ◇ otherwise we don't know what happened
- ◇ how was it calculated? is there a mistake? who knows?
- ◇ pol sci perspective
[*]<http://gking.harvard.edu/files/gking/files/replication.pdf>
- ◇ IT perspective [*]<http://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.1001745>

rules for everyday practice [revisit/stress later!!]

- ◇ once you have coded everything, double check
 - leave it aside (2wks) and check again (does wonders)
 - show it to other people, post on your website
- ◇ cross-check end output with raw data
 - eg same values for couple data points? do values make sense?
- ◇ check with alt data? they tell the same story?
 - I Google/Gemini values/tables/graphs from my output
- ◇ everything has been already studied by others
- ◇ like lit rev, its data rev

outline

the idea

get code from others!

the easiest way to do research in 21st century

- ◇ start with code others wrote, and build on their work
- ◇ fastest most efficient way to do research
- ◇ just email author and ask for code
- ◇ even if sas, stata, etc—easy to figure out what's going on and do sth something similar in Py